Experiment No.7
Implementation of User Registration and Session-Based Login Using PHP and MySQL
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Aim: To implement user registration and session-based login using PHP and MySQL.

Objective: To design a user registration system, store user data securely in a MySQL database, and implement session-based authentication to manage user login and access control.

Theory:

A user registration system is an essential part of web applications that allows users to create accounts and securely store their information. Using PHP, developers can design forms to collect user details such as name, email, and password, and process this data on the server side. MySQL is used as the database to store and manage user information efficiently, ensuring data is organized and easily retrievable.

To ensure security, user input is validated and sensitive information like passwords is stored in an encrypted format. Session-based authentication is implemented using PHP sessions, which allows the server to maintain user login state across multiple pages. This helps in controlling access to protected areas of the application and ensures that only authorized users can access certain features.

Implementation:

Dashboard.php

```
<?php
session_start();

if (!isset($_SESSION['user_id'])) {
    header("Location: login.php");
    exit();
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Dashboard</title>
```

```
</head>
<body>
    <h2>Welcome to Dashboard</h2>

    <p><strong>User ID:</strong> <?php echo $_SESSION['user_id']; ?></p>
    <p><strong>Name:</strong> <?php echo $_SESSION['full_name']; ?></p>
    <p><strong>Email:</strong> <?php echo $_SESSION['email']; ?></p>

    <br>
    <a href="logout.php">Logout</a>
</body>
</html>
```

Db.php

```
<?php
$host = "127.0.0.1";
$username = "root";
$password = "";   // or your actual MySQL password
$database = "user_system_db";
$port = 3307;

$conn = new mysqli($host, $username, $password, $database, $port);

if ($conn->connect_error) {
    die("Connection failed: " . $conn->connect_error);
}
?>
```

Login_process.php

```
<?php
session_start();
include 'db.php';

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $email = trim($_POST['email']);
    $password = $_POST['password'];

    $sql = "SELECT id, full_name, password FROM users WHERE email = ?";
    $stmt = $conn->prepare($sql);
    $stmt->bind_param("s", $email);
    $stmt->execute();
    $result = $stmt->get_result();

    if ($result->num_rows == 1) {
        $user = $result->fetch_assoc();

        if (password_verify($password, $user['password'])) {
            $_SESSION['user_id'] = $user['id'];
            $_SESSION['full_name'] = $user['full_name'];
            $_SESSION['email'] = $email;

            header("Location: dashboard.php");
            exit();
        } else {
```

```
            echo "Invalid password. <a href='login.php'>Try again</a>";
        }
    } else {
        echo "User not found. <a href='login.php'>Try again</a>";
    }

    $stmt->close();
    $conn->close();
}
?>
```

Login.php

```
<?php
session_start();

if (isset($_SESSION['user_id'])) {
    header("Location: dashboard.php");
    exit();
}
?>

<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Login</title>
</head>
<body>
    <h2>User Login</h2>

    <form action="login_process.php" method="POST">
        <label>Email:</label><br>
        <input type="email" name="email" required><br><br>

        <label>Password:</label><br>
        <input type="password" name="password" required><br><br>

        <button type="submit">Login</button>
    </form>

    <p>Don't have an account? <a href="register.php">Register here</a></p>
</body>
</html>
```

Logout.php

```
<?php
session_start();
session_unset();
session_destroy();

header("Location: login.php");
exit();
?>
```

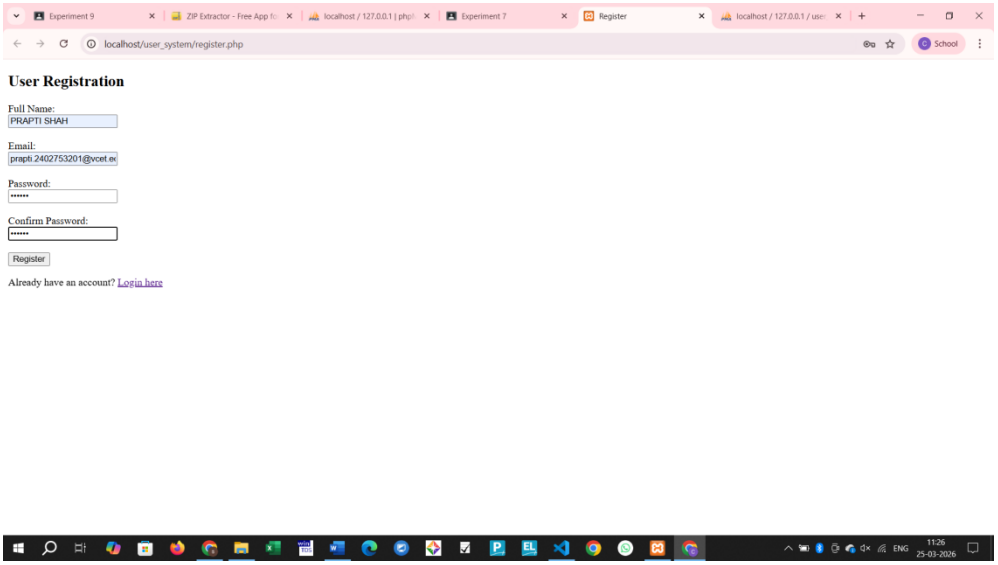
Register_Process.php

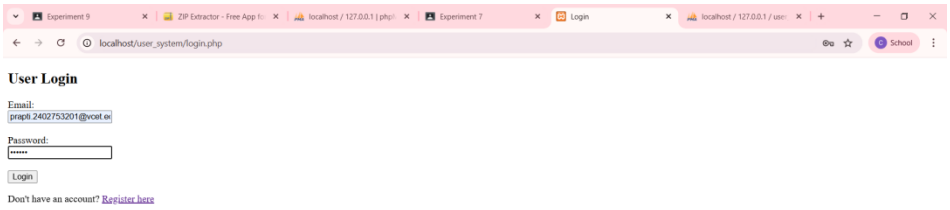
```php
<?php
include 'db.php';

if ($_SERVER["REQUEST_METHOD"] == "POST") {
    $full_name = trim($_POST['full_name']);
    $email = trim($_POST['email']);
    $password = $_POST['password'];
    $confirm_password = $_POST['confirm_password'];

    if ($password !== $confirm_password) {
        die("Passwords do not match.");
    }

    // Check if email already exists
    $check_sql = "SELECT id FROM users WHERE email = ?";
    $stmt = $conn->prepare($check_sql);
    $stmt->bind_param("s", $email);
    $stmt->execute();
    $stmt->store_result();

    if ($stmt->num_rows > 0) {
        die("Email already registered.");
    }
    $stmt->close();

    // Hash password
    $hashed_password = password_hash($password, PASSWORD_DEFAULT);

    // Insert user
    $sql = "INSERT INTO users (full_name, email, password) VALUES (?, ?, ?)";
    $stmt = $conn->prepare($sql);
    $stmt->bind_param("sss", $full_name, $email, $hashed_password);

    if ($stmt->execute()) {
        echo "Registration successful. <a href='login.php'>Login here</a>";
    } else {
        echo "Error: " . $conn->error;
    }

    $stmt->close();
    $conn->close();
}
?>
```

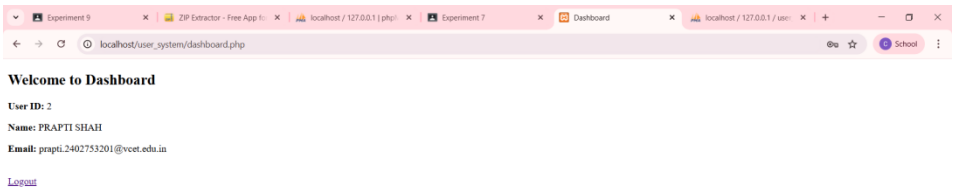
Register.php

```html
<!DOCTYPE html>
<html lang="en">
<head>
    <meta charset="UTF-8">
```

```
    <meta name="viewport" content="width=device-width, initial-scale=1.0">
    <title>Register</title>
</head>
<body>
    <h2>User Registration</h2>

    <form action="register_process.php" method="POST">
        <label>Full Name:</label><br>
        <input type="text" name="full_name" required><br><br>

        <label>Email:</label><br>
        <input type="email" name="email" required><br><br>

        <label>Password:</label><br>
        <input type="password" name="password" required><br><br>

        <label>Confirm Password:</label><br>
        <input type="password" name="confirm_password" required><br><br>

        <button type="submit">Register</button>
    </form>

    <p>Already have an account? <a href="login.php">Login here</a></p>
</body>
</html>
```







Conclusion: Write about designing a user registration system, storing user data securely in a MySQL database, and implementing session-based authentication to manage user login and access control.

A user registration system is designed to allow new users to create an account by entering details such as name, email, username, and password. During registration, the system first validates the input to ensure that all required fields are filled correctly, the email format is valid, and the username or email is not already registered. After validation, the password should never be stored in plain text. Instead, it must be converted into a secure hashed form using a strong hashing algorithm such as bcrypt before saving it in the MySQL database. This helps protect user credentials even if the database is accessed by unauthorized persons.

To store user data securely in MySQL, a properly structured table is created with fields like user_id, name, email, username, password_hash, created_at, and role. Sensitive operations should use prepared statements or parameterized queries to prevent SQL injection attacks. Additional security measures such as unique constraints on email and username, input sanitization, and limited database privileges further improve protection. Password reset options, timestamp tracking, and role-based fields can also be included to make the system more robust and manageable.

Session-based authentication is used to manage user login and access control after registration. When a user logs in, the entered username or email and password are checked against the stored database record. If the password matches the hashed password, the server creates a session and stores important information such as the user's ID, username, and role in that session. A session ID is then sent to the browser, usually through a secure cookie. On each new request, the system verifies the session to confirm whether the user is authenticated. This allows only logged-in users to access protected pages such as dashboards, profile settings, or admin panels.

Access control is implemented by checking the session before displaying restricted resources. If no valid session exists, the user is redirected to the login page. If role-based access is required, the system checks the user's stored role in the session to decide whether they can view normal user pages or admin-only sections. Logout is handled by destroying the session and clearing the cookie, ensuring that the user's authenticated state is removed. Overall, this approach creates a secure and organized registration and login system that protects user data and controls access effectively.

MDC401: Introduction to Web Technologies